

Types of Chunking, Embedding, and Retrieval

1. Chunking Types

Chunking Type	Definition	Method / Class
Fixed-size Chunking	Splits text into equal-size chunks.	RecursiveCharacterTextSplitter
Recursive Chunking	Splits using paragraph → sentence → word.	RecursiveCharacterTextSplitter
Token-based Chunking	Splits based on token counts.	TokenTextSplitter
Semantic Chunking	Splits based on meaning.	SemanticChunker
Sentence-based Chunking	Splits by sentence.	NLTKTextSplitter / SpacyTextSplitter
Document-based Chunking	Splits using headers or sections.	MarkdownHeaderTextSplitter

2. Embedding Types

Embedding Type	Definition	Method / Class
Dense	High-dimensional semantic vectors.	HuggingFaceEmbeddings / OpenAIEmbeddings
Sparse (BM25)	Keyword-based retrieval vectors.	BM25Retriever
Hybrid	Combines dense + sparse.	EnsembleRetriever
Multi-vector	Multiple embeddings per chunk.	MultiVectorRetriever
Long-context	Designed for large docs.	VoyageEmbeddings

3. Retrieval Types

Retrieval Type	Definition	Method / Class
Vector Search	Finds nearest embeddings.	FAISS / Chroma / Pinecone
BM25	Keyword-based scoring.	BM25Retriever
Hybrid	Dense + sparse retrieval.	EnsembleRetriever
Contextual Compression	Compresses results.	ContextualCompressionRetriever
Parent Document	Returns full parent doc.	ParentDocumentRetriever
Multi-Query	Creates multiple queries.	MultiQueryRetriever
Self-Query	Adds filters via LLM.	SelfQueryRetriever
Time-based	Retrieval by timestamps.	TimeWeightedVectorStoreRetriever